

InsureFlow – Insurance Management Platform Documentation

Introduction

InsureFlow is a full-stack Insurance Management System developed to simplify insurance policy management through a secure and user-friendly digital platform. The application enables customers to purchase insurance policies, make premium payments, submit insurance claims, and monitor their insurance activities through a centralized dashboard. Administrators can manage insurance plans, review customer claims, and oversee the overall system.

The project demonstrates enterprise-level backend development using **Java**, **Spring Boot**, **Spring Security**, **JWT Authentication**, **PostgreSQL**, and a modern **React** frontend. The application follows a layered architecture that emphasizes scalability, maintainability, and security while implementing industry-standard RESTful APIs.

Design Inspiration

This project is inspired by modern insurance management platforms such as:

1. PolicyBazaar

- Online insurance management
- Digital policy purchase
- Premium comparison
- Customer dashboard

2. ACKO Insurance

- Digital-first insurance experience
- Simplified claim process
- Online premium payments
- Customer-centric workflow

3. InsureFlow Custom Implementation

- JWT Authentication
 - Policy Management
 - Premium Payment System
 - Insurance Claim Management
 - Role-Based Authorization
 - PostgreSQL Database
 - React Dashboard
 - Spring Boot REST APIs
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Roles

Customer

- Register account
 - Login securely
 - Browse insurance plans
 - Purchase insurance policies
 - View purchased policies
 - Pay insurance premiums
 - Submit insurance claims
 - Track claim status
 - View payment history
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Administrator

- Login securely
 - Create insurance plans
 - Update insurance plans
 - Delete insurance plans
 - View all purchased policies
 - View all premium payments
 - View all claims
 - Approve or reject insurance claims
 - Manage insurance products
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Agent (*Future Enhancement*)

- Login securely
 - View assigned customers
 - Review claims
 - Assist customers
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Use Cases of the Project

Key Use Cases

User Registration

- Customer creates an account
- Password encrypted using BCrypt
- User stored securely in PostgreSQL

User Authentication

- Customer logs in
- JWT token generated
- Protected APIs accessed using Authorization Bearer Token

Premium Payment

- Select purchased policy
- Make premium payment
- Store payment history
- Generate transaction details

Claim Management

- Submit insurance claims
- Enter claim amount
- Provide reason and description
- Track claim status

Administration

- Manage insurance plans
 - Review customer claims
 - Approve claims
 - Reject claims
 - View complete system information
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Tech Stack

Backend

Spring Boot

- REST API development
- Dependency Injection
- Layered Architecture
- Business Logic Implementation

Spring Security

- Endpoint protection
- Authentication management
- Authorization handling
- Secure API access

JWT (JSON Web Token)

- Stateless authentication
- Secure API communication
- Authorization header validation
- Token-based access control

Spring Data JPA (Hibernate)

- ORM implementation
- Repository abstraction
- Entity management
- Database operations

PostgreSQL

- Users
- Policy Types
- Policies
- Premium Payments
- Claims

BCrypt Password Encoder

- Password Hashing
 - Secure Password Storage
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Frontend

React

- Component-based architecture
- Dynamic UI rendering
- API integration
- State management

Tailwind CSS

- Utility-first styling
- Responsive layouts
- Modern dashboard interface
- Faster UI development

Axios

- REST API communication
 - JWT header injection
 - Request interceptors
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Authentication

Authentication is implemented using JWT (JSON Web Token).

Flow

User Login



Credential Validation



JWT Token Generated



Authorization: Bearer <TOKEN>



Protected API Access

Security Layers

JwtFilter

- Validates JWT tokens
- Extracts authenticated users
- Secures protected endpoints

Spring Security

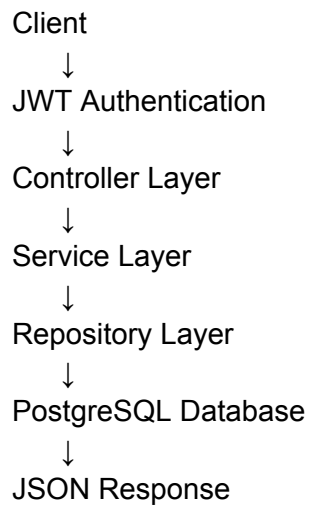
- Authorization management
- Endpoint security
- Authentication context

BCrypt Password Encoder

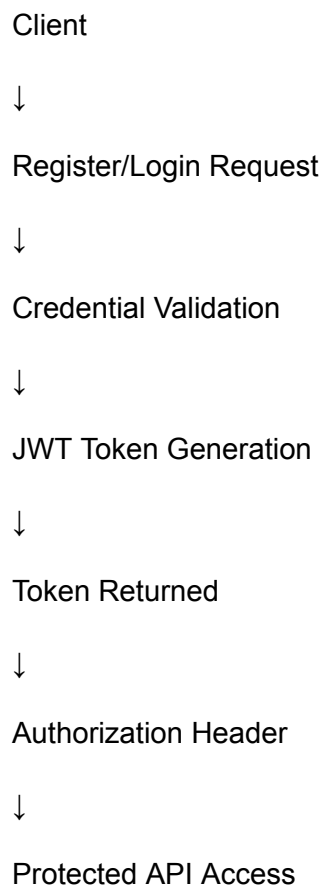
- Password hashing
 - Secure credential storage
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Flowcharts

Backend Flow



Authentication Flow



Policy Purchase Flow

Customer



Browse Policy Types



Select Policy



Purchase Policy



Database Storage



Policy Created

Premium Payment Flow

Customer



Select Purchased Policy



Choose Payment Method



Payment Processing



Payment Stored



Payment History Updated

Claim Flow

Customer



Select Policy



Enter Claim Details



Submit Claim



Database Storage



Status = Pending



Admin Review



Approve / Reject

System Architecture

React Frontend



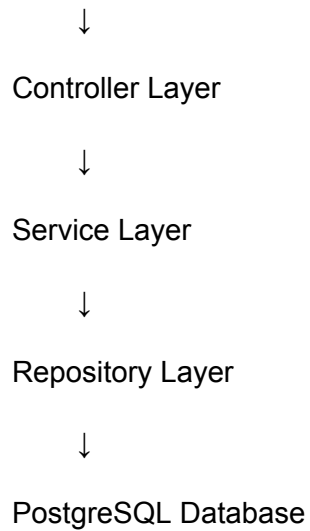
REST API



Spring Security

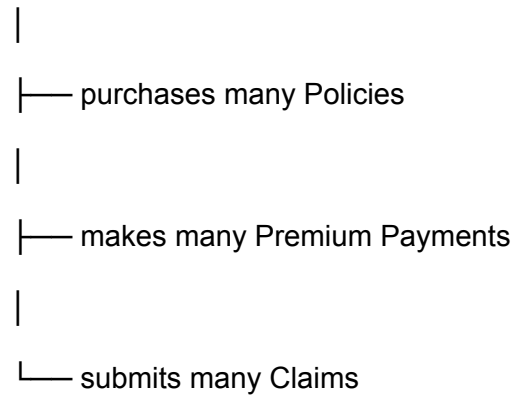


JWT Authentication

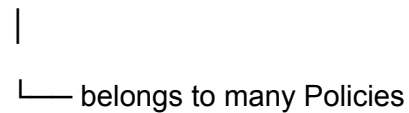


Database Design (ER Concept)

User



PolicyType



Policy



|— belongs to one User
|
|— belongs to one PolicyType
|
└— has many Premium Payments

Claim

|
└— belongs to one Policy

PremiumPayment

|
└— belongs to one Policy

Frontend Integration Flow

React Dashboard



REST API Calls



Spring Boot Backend



JWT Authentication



PostgreSQL Database



JSON Response



Dashboard Update

Industry Value of the Project

This project demonstrates real-world enterprise software development concepts.

Industry Applications

- Insurance Companies
 - Policy Management Systems
 - Digital Insurance Platforms
 - Banking Insurance Products
 - Financial Service Providers
 - Enterprise Insurance Portals
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Key Concepts Implemented

- JWT Authentication
 - Spring Security
 - RESTful API Development
 - Layered Architecture
 - DTO-Based Design
 - CRUD Operations
 - Role-Based Authorization
 - Global Exception Handling
 - Validation
 - PostgreSQL Database Design
 - Full-Stack Development
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Technologies Used

Technologies Used

Backend

- Java 21
 - Spring Boot
 - Spring Security
 - JWT
 - Spring Data JPA
 - Hibernate
 - PostgreSQL
 - Maven
 - Lombok
 - BCrypt Password Encoder
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Frontend

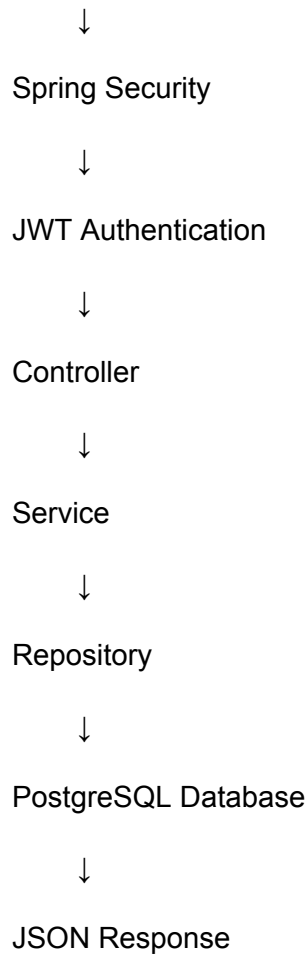
- React
 - Tailwind CSS
 - Fetch API / Axios
 - Vite
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Integration Overview

React Frontend



REST API



Conclusion

InsureFlow successfully demonstrates the design and implementation of a secure, scalable, and enterprise-ready Insurance Management System.

Key Achievements

- Secure JWT Authentication
- Role-Based Authorization
- Insurance Policy Management
- Premium Payment System
- Insurance Claim Management
- PostgreSQL Relational Database
- Global Exception Handling
- Request Validation
- RESTful API Development
- Layered Software Architecture
- React Dashboard Integration

The project draws inspiration from modern digital insurance platforms such as **PolicyBazaar** and **ACKO Insurance**, while introducing a custom implementation focused on secure authentication, policy lifecycle management, premium payment processing, and claim handling.

InsureFlow showcases industry-standard backend development practices using **Spring Boot, Spring Security, JWT Authentication, PostgreSQL, React, REST APIs, DTO-based architecture, and role-based access control**, making it a practical, scalable, and portfolio-worthy full-stack application.